

① a) 0 1 1 0 0 1 1 1 2

$$\begin{aligned}
 & \rightarrow 1 \cdot 2^0 = 1 \\
 & \rightarrow 1 \cdot 2^1 = 2 \\
 & \rightarrow 1 \cdot 2^2 = 4 \\
 & \rightarrow 0 \cdot 2^3 = 0 \\
 & \rightarrow 0 \cdot 2^4 = 0 \\
 & \rightarrow 1 \cdot 2^5 = 32 \\
 & \rightarrow 1 \cdot 2^6 = 64 \\
 & \rightarrow 0 \cdot 2^7 = 0 \quad / \quad 103_{10}
 \end{aligned}$$

b) 110010110100001112

1 9 6 8 7 16

c) 82_{10} $\cancel{28} \text{ (64) } \cancel{32} \text{ (16) } \cancel{8} \text{ (4) } \cancel{2}$ $82 - 64 = 18$

01010010_2 $18 - 16 = 2$

d) $0 \times 2^7 + 6 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 5 \times 2^2 + 96 + 0$

$10 \cdot 16^3 + 15 \cdot 16^2$

$40960 + 3840 + 96 + 0 = 44896_{10}$

e) $0 \times 2^8 + 10 \times 2^7 + 1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$1010101100100001111_2 = 1010101100100001111_2$

f) 912342_{10}

para fazer calculo dividido sucessivas por 16, e resto é o número convertido

$912342 / 16 = 57021$	rests 6 - 6	$0 \times 9 e 8 2 6 1 6$
$57021 / 16 = 3563$	rests 13 - 7	
$3563 / 16 = 222$	rests 11 - 3	
$222 / 16 = 13$	rests 14 - e	
$13 / 16 = 0$	rests 13 - 7	

9) 131_{10}

(128) $6/4 \quad 3/2 \quad 1/6 \quad 8/8 \quad 4/4 \quad (2)(1)$

10000011_2

2

a) $1 \text{ Gib} = 1024^3 = 1073741824 \text{ bytes}$

$$\frac{1073741824}{1073741824} = 1 \text{ Gib}$$

b) $1 \text{ GB} = 1000^3 = 1000000000 \text{ bytes}$

$$\frac{1000000000}{1000000000} = 1 \text{ GB}$$

c) $1 \text{ KB} = 1000 \text{ bytes}$

$132 \cdot 1000 = 132000 \text{ bytes}$

d) $12 \cdot 1000^3 = 12000000000$

$$\frac{12000000000}{1073741824} = 11,18 \text{ Gib}$$

$$\textcircled{2} \quad 132 \text{ mB} = 132 \ 000 \ 000 \text{ bytes}$$

$$\frac{132 \ 000 \ 000}{1024^2} = 125,89 \text{ MB}$$

$$\textcircled{1} \quad \text{mB} - \text{megabits}$$

$$8 \text{ bits} = 1 \text{ byte}$$

$$11512 / 8 = 1439 \text{ MB}$$

$$\frac{1439}{1000} = 1,439 \text{ GB}$$

$$\textcircled{3} \quad 4 \text{ bytes} = 32 \text{ bits}$$

$$2^{32} - 1 = 4 \ 294 \ 967 \ 295$$

$$\textcircled{4} \quad 1 \text{ byte} = 8 \text{ bits}$$

$$40 / 8 = 5 \text{ bytes}$$

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Operações binárias

$$\textcircled{a} \quad \begin{array}{r} 111011101 \\ + 11010010 \\ \hline \end{array}$$

$$110101112$$

$$\textcircled{b} \quad \begin{array}{r} 10110101 \\ - 00100110 \\ \hline \end{array}$$

$$10001112$$

$$\textcircled{c} \quad \begin{array}{r} 101111 \\ \times 11 \\ \hline \end{array}$$

$$101111$$

$$101111 +$$

$$100011012$$

Represente os valores em complemento a dois

a) $\overset{\text{MSB}}{\uparrow}$
 $(a) \quad 11011000_2$

00100111
 $\hookrightarrow \text{some 1}$

$0010100 \quad -40_{10}$
 $\hookrightarrow 32+8=40$

b) $\overset{\text{MSB}}{\uparrow}$
 $(b) \quad 00011101_2$

29_{10}

c) $\overset{\text{MSB}}{\uparrow}$
 $(c) \quad 1101110_2$

$00100001_2 \rightarrow \text{some 1}$

0010010_2
 $32+2=34 \quad \rightarrow -34_{10}$

Complementos a 1

a) $4_{10} + 18_{10}$

$00000100 \quad 22_{10}$
 $+ \quad 00010010$

 00010110

b) $118 - 14$

some
 0110110
 $+ \quad 11110001$

 10110011
 $\hookrightarrow 1$

 001101000_2

118: 0110110

14: 00001110

C: 11110001

c) $-58 + 49$

58: 00111010

C: 1100101

49: 00110001

same

1100101

00110001

1110110

invert

00001001

-9_{10}

d) $102 - 12$

102: 01100110

12: 00001100

C: 11110011

same

01100110

+ 11110011

101011001

101011001

01011010

90_{10}